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Supervised Learning

08 July 2026

What is Supervised learning?

Definition: Supervised Learning is a type of machine learning where an algorithm learns from labeled data.

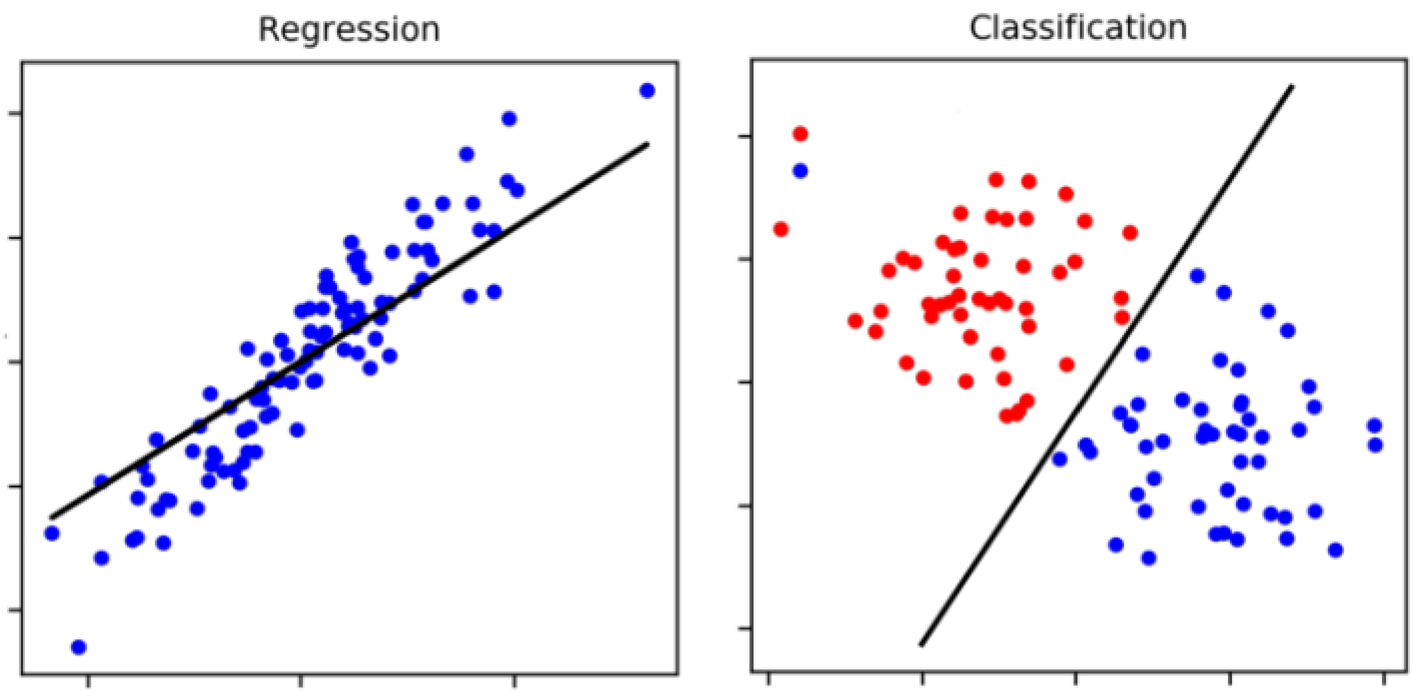
Each training example includes input (features) and the correct output (label).

Key Ideas:

- The algorithm tries to learn a function that maps inputs (X) to outputs (y).
- Once trained, the model can predict outputs for new, unseen inputs.

Supervised learning types

Regression vs classification =
continuous number vs category/class label



Regression Examples

Regression examples:

- predicting number of people who will click a Google ad based on the ad content and data about the user's prior online behavior,
- predicting the number of traffic accidents based on road conditions and speed limit,
- predicting weather parameters (such as wind speed) based on historical weather behaviour.

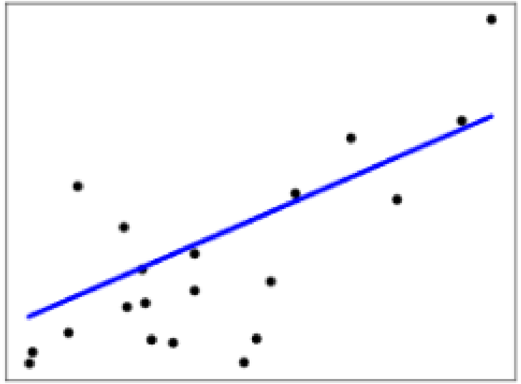
Regression predictive modeling is the task of approximating a mapping function (f) from input variables (X) to a continuous output variable (y).

y = f(X), X = input features, y = target variable

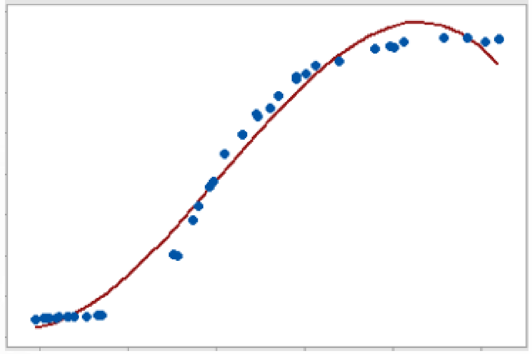
Regression Algorithms

Optimal regression model depend on dataset

- Linear



- Non-Linear



Regression Algorithms - Linear

- Simple linear regression
- Ordinary Least Squares

  scikit learn `LinearRegression()`

- Stochastic Gradient Descent

  `SGDRegressor(max_iter=1000, tol=1e-3)`

- Support Vector Regression (kernel = linear)

Please note, this is not a comprehensive list, there are many more algorithms that may suite you purpose, we are simply listing ***some* of the algorithms available in python.

Regression Algorithms – Non-Linear

Some examples of (ensemble) algorithms:

- Decision Trees
- Random Forest Regression (https://scikit-learn.org/stable/modules/generated/sklearn.ensemble.RandomForestRegressor.html)
- XGBoost (https://www.geeksforgeeks.org/machine-learning/xgboost/)
- LightGBM (https://www.geeksforgeeks.org/machine-learning/lightgbm-light-gradient-boosting-machine/)
- CatBoost (https://www.geeksforgeeks.org/machine-learning/regression-using-catboost/)

Regression Algorithms – Non-Linear

Some (more) examples of algorithms:

- K-Nearest Neighbors (KNN) (https://www.geeksforgeeks.org/machine-learning/k-nearest-neighbours/)
- Support vector regression (kernel = polynomial or rbf) (https://scikit-learn.org/stable/modules/generated/sklearn.svm.SVR.html)
- Kernel ridge regression (https://www.geeksforgeeks.org/machine-learning/understanding-kernel-ridge-regression-with-sklearn/)
- Multi layer perceptron (Deep learning) (https://www.datacamp.com/tutorial/multilayer-perceptrons-in-machine-learning)
- Artificial Neural network (LSTM/ Recurrent Neural Network) (Deep learning)

Regression – Error metrics

There are three error metrics that are commonly used for evaluating and reporting the performance of a regression model

- Mean Squared Error (MSE)
- Root Mean Squared Error (RMSE)
- Mean Absolute Error (MAE)

$$\mathrm{MSE} = \frac{1}{n}\sum_{i=1}^{n}(Y_i - \hat{Y}_i)^2$$

$$RMSE = \sqrt{\frac{\sum_{i=1}^{N}\|y(i) - \hat{y}(i)\|^2}{N}}$$

$$\mathrm{MAE} = \frac{\sum_{i=1}^{n}|y_i - x_i|}{n}$$

N / n = total number of samples

$\hat{y}$ / x_i = true value

y_i = prediction

Classification Examples

Classification is used when the target, or value to predict, is a discrete class label.

Classification examples:

- Spam filtering
- Identifying an object in an image
- Customer behaviour prediction

Classification Algorithms

Some examples of algorithms:

- Logistic Regression (https://scikit-learn.org/stable/modules/generated/sklearn.linear_model.LogisticRegression.html)
- Decision trees (https://scikit-learn.org/stable/modules/tree.html)
- Random Forest (https://scikit-learn.org/stable/modules/generated/sklearn.ensemble.RandomForestClassifier.html)
- XGBoost
- LightGBM
- CatBoost

Classification Algorithms

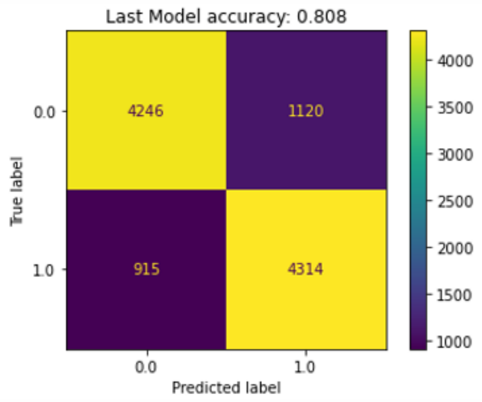
Some examples of algorithms:

- Support Vector Machines (https://scikit-learn.org/stable/modules/svm.html)
- Naive Bayes (https://medium.com/low-code-for-advanced-data-science/an-introduction-to-the-naive-bayes-classifier-and-how-to-use-it-bba78826ef8f)
- k- Nearest Neighbors
- Multi layer perceptron
- Artificial neural networks (e.g., Convolutional neural networks)

Classification - Error metrics

There are several error metrices that can be used for classifiers.

- Accuracy – how often the classifier correctly predicts the target.
- Recall – number of correct positive predictions made from all positive predictions that could be made.
- Precision – quantifies the proportion of positive identifications that was correct.
- F1 score – combined idea of precision and recall. Max when precision is equal to recall – best for unbalanced datasets.
- Confusion matrix



Preparing the data

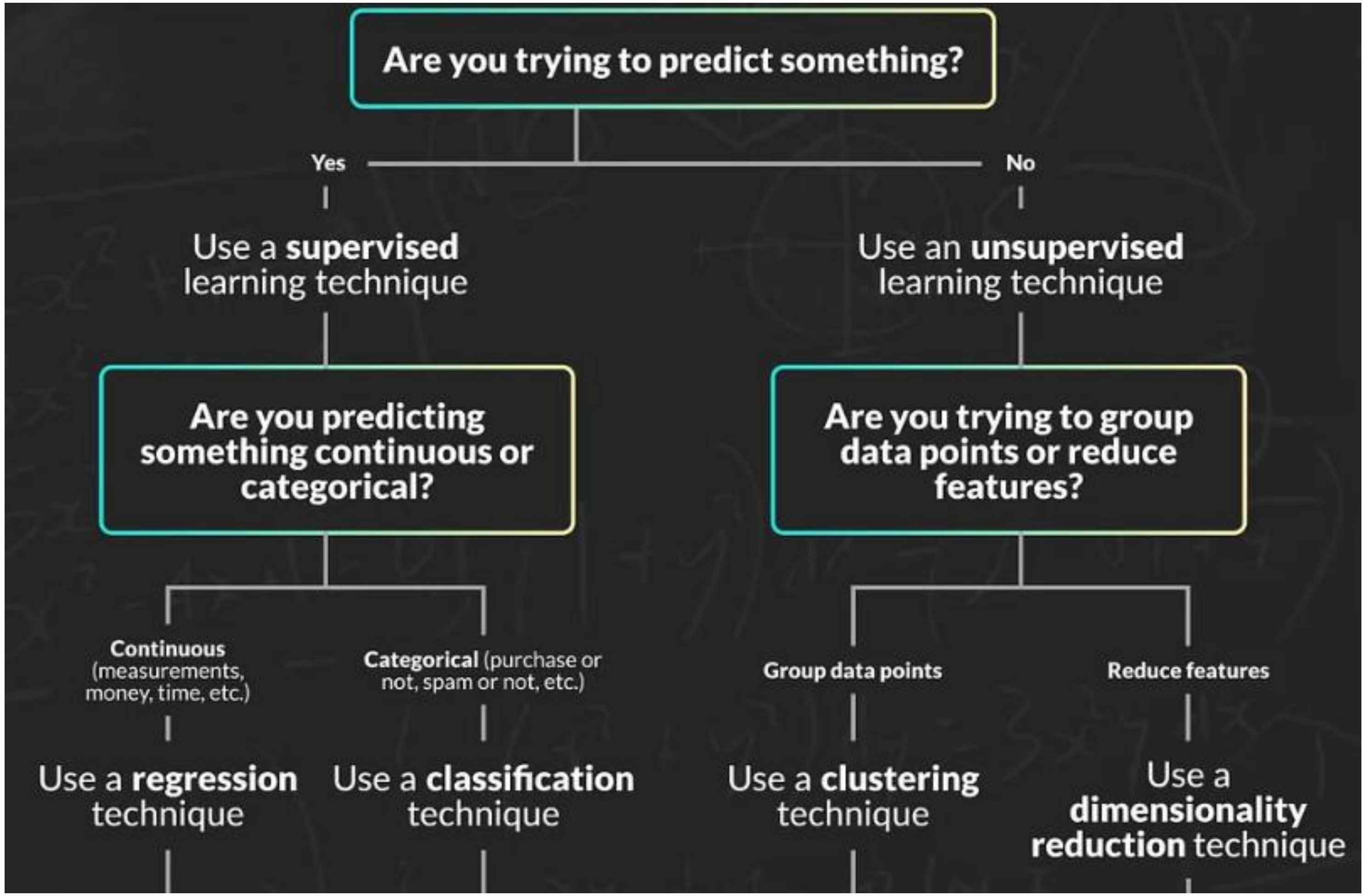
- EDA (exploratory data analysis)
- Preparing Data
  - Rescale Inputs (normalization/standardization)
  - Remove Collinearity (if exist)
  - Handle multiple (one-hot encoding) or imbalances classes (over sampling / under sampling / SMOTE - https://machinelearningmastery.com/smote-oversampling-for-imbalanced-classification/)
- Additional operations if needed (applicable to **Linear regression**)
  - Linear Assumption
  - Remove Noise
- Dimensionality Reduction (many features (X)) – PCA (https://www.geeksforgeeks.org/data-analysis/principal-component-analysis-with-python/)

Workflow

- ✓ Prepare/create dataset
- ❑ Select the model
- ❑ Build the model
- ❑ Train the model
- ❑ Make predictions
- ❑ Communicate results

How to choose a model

Medium: https://medium.com/learning-data/which-machine-learning-algorithm-should-i-use-for-my-analysis-962aeff11102



Workflow

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Building the Model

Preparing Data For Regression or Classification

- Randomisation
- Test/Train split

```
from sklearn.model_selection import train_test_split
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=1)
```

model = XGBClassifier(**params)

Workflow

- ✓ Prepare/create dataset
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- ❑ Communicate results

Training the Model

model.fit(X_train, y_train)

Cross validation (https://scikit-learn.org/stable/modules/cross_validation.html)

Hyper parameter tuning (manual / grid search / random search)
(https://www.geeksforgeeks.org/machine-learning/hyperparameter-tuning/)

Workflow

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Predict and Evaluate

y_pred = model.predict(X_test)

```
accuracy = accuracy_score(y_test, y_pred)
print("Model Accuracy:", accuracy)

print("\nClassification Report")
print(classification_report(y_test, y_pred))
```

Output:

```
Model Accuracy: 0.9090909090909091

Classification Report
              precision    recall  f1-score   support

           0       0.80      0.92      0.85        38
           1       0.97      0.90      0.93        94

    accuracy                           0.91       132
   macro avg       0.88      0.91      0.89       132
weighted avg       0.92      0.91      0.91       132
```

Source: https://www.geeksforgeeks.org/machine-learning/xgboost/

Workflow

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Results Communication / Sharing

Type of communication is audience dependent

- Science communication
- Dashboard
- App
- API

Workflow

- ✓ Prepare/create dataset
- ✓ Select the model
- ✓ Build the model
- ✓ Train the model
- ✓ Make predictions
- ✓ Communicate results

Additional

Transfer Learning

- Fine-Tuned Transformers - general-purpose model (pretrained) and retraining it on a specialized dataset (https://medium.com/@hassaanidrees7/fine-tuning-transformers-techniques-for-improving-model-performance-4b4353e8ba93)
  (e.g., RoBERTa, DeBERTa, HateBERT, BERTweet)

- YOLOv12 (https://docs.ultralytics.com/models/yolo12#usage-examples)
- ResNet (https://www.geeksforgeeks.org/deep-learning/residual-networks-resnet-deep-learning/)

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